

A Machine Learning Approach to Improving Food Insecurity Measurement Using Survey Data

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1. Introduction and Motivation

1.1. Food Insecurity in the U.S. and New York

Food insecurity, defined as the inability to consistently access safe, healthy, affordable, and culturally appropriate food, is a profound health, economic, and social challenge afflicting over 18 million households in the U.S.ⁱ In New York City, an estimated 40% of adults live in a household at risk of being food insecure.ⁱⁱ Individuals experiencing food insecurity face a higher risk of malnutrition, chronic diseases, and mental health challenges.ⁱⁱⁱ This lack of consistent access to adequate food can also impair physical and mental work capacity, hindering the economic and societal potential of millions. Moreover, childhood food insecurity and resulting malnutrition can lead to permanent developmental issues.

Poverty, income insecurity, and high living costs are the main drivers of food insecurity. Community factors, including under-resourced schools, poor transportation, and unsafe neighborhoods, further compound the problem.^{iv}

To combat food insecurity, federal, state, and local governments support and promote a variety of food assistance programs. The Supplemental Nutrition Assistance Program (SNAP), also known as food stamps, accounts for 68% of US Department of Agriculture (USDA) nutrition assistance spending and serves an average of 42 million participants monthly.^v Other crucial programs include the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), school meal programs, and emergency food assistance initiatives.^{vi} Despite their importance, these benefits often fall short of adequately addressing the need, may not allow for the purchase of healthy food, or face issues of accessibility and underutilization by those who could benefit most..^{vii}

1.2. Research Questions

Timely and accurate identification of food-insecure populations is essential for designing effective interventions and allocating resources equitably and efficiently. Yet existing survey tools often fall short of providing the geographic and demographic granularity needed to inform local policy decisions.

This study investigates how machine learning methods can enhance the measurement of food insecurity by leveraging existing survey data. Specifically, it seeks to answer the following research questions:

- I. Can a machine learning model, trained on existing survey data, accurately predict which households are food insecure in New York State?
- II. Which specific survey questions are most informative in identifying food insecure households?

To address these questions, this project relies on two data sources: the **Current Population Survey Food Security Supplement (FSS)** and the **American Community Survey (ACS)**. The FSS, administered annually by the USDA, provides detailed data on food access, adequacy, and related household behaviors for approximately 54,000 U.S. households.^{viii} Official statistics on food insecurity in the U.S. are primarily derived from the data collected by the FSS. The ACS is a much larger survey that collects demographic, housing, and economic data on roughly 3.5 million households each year.^{ix}

The first research question involves building a random forest classifier using the FSS data, trained on household and demographic variables that are also available in the ACS. The goal is to create a predictive model that can be applied to the ACS, enabling more geographically granular estimates of food insecurity than the FSS alone allows. However, as the results will show, models using only these shared covariates struggle to accurately identify food-insecure households.

The second research question explores how the predictive power of this model changes when individual FSS survey questions—those used to compute the official food security score—are added as features. By evaluating which questions most improve the model’s **recall** for food-insecure households, this analysis identifies which questions would be most valuable to include in the ACS to enhance its capacity to measure food insecurity. This approach offers a data-driven basis for potential future survey integration.

2. Data Description

2.1. Current Population Survey Food Security Supplement (FSS)

This report relies primarily on data from the Current Population Survey Food Security Supplement (FSS) covering 2019 – 2023.^x The FSS is a household-level survey representative of the U.S. population. It asks detailed food security questions, including weekly spending on food, whether the household regularly runs out of food, and whether the household can afford balanced meals. Based on responses to these questions, the surveyors categorize each household into one of three categories: food secure, low food security, or very low food security.¹ This food security classification is the primary variable of interest in this study. The FSS also collects demographic and household characteristics, including household size and income, race, educational attainment, and age of household members. Geographic information, including state and county, is also recorded.

Annually, the FSS surveys approximately 54,000 households across the U.S., 1,000 of which are in New York State.

¹ For more information, see Matthew Rabbitt et al., “[Household Food Security in the United States in 2022](#),” 2023, page 5.

2.2. American Community Survey (ACS)

The American Community Survey (ACS) is a significantly larger household-level survey covering the U.S. population.^{xi} The ACS gathers a broad range of population, housing, and household characteristics at the state, county, and census tract level.^{xii} Notably, the ACS does not collect data on food security. However, the ACS annually surveys 3.5 million households nationwide and 100,000 households within New York State. Furthermore, the ACS provides geographic identification at the Public Use Microdata Area (PUMA) level. Each PUMA has a population between 100,000 and 200,000 residents; thus, in densely populated counties, PUMA-level statistics can provide more granular estimates of sub-state indicators and trends.^{xiii}

3. Methods

This study relies upon the FSS dataset to train a **random forest classification** model. Random forests are a supervised machine learning algorithm that use labeled data to identify underlying patterns and classify unlabeled data based on common covariates, also known as “features.”^{xiv} A random forest aggregates predictions from numerous individual decision trees, each trained on random subsets of data and features, and chooses the most popular prediction for each observation.^{xv}

In this report, the target variable (label) is the **food security classification**; the labeled training data is the FSS dataset; and the unlabeled data for prediction is the ACS dataset. The **baseline model** is constructed using covariates common to both the FSS and the ACS. This baseline model can then be applied to the ACS data to generate predictions regarding the food security status of each household. Subsequently, **extension models** are developed by incorporating specific FSS survey questions—those used to derive the food security classification—as additional features. The results from these extension models aim to identify which of these survey questions exhibit the strongest predictive capability.

3.1. Label: Standard and Binary Food Security Classification

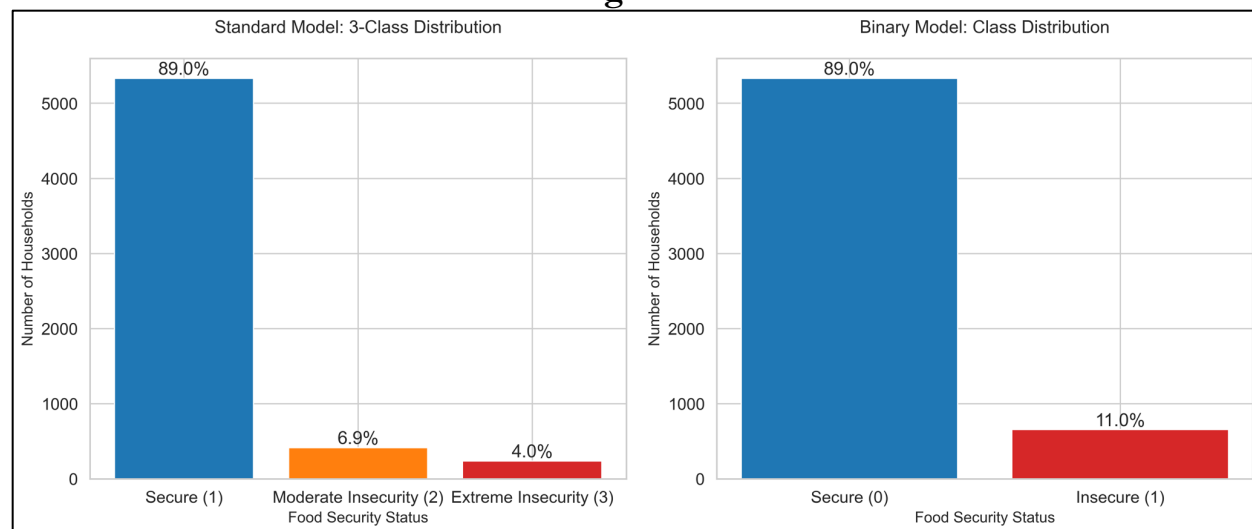
As previously mentioned, the FSS classifies each household as (1) food secure, (2) low food security, or (3) very low food security. In addition to this standard classification, this analysis also explores a binary classification. In the binary approach, households classified as low food security or very low food security in the standard classification are grouped as ‘food insecure,’ while those classified as food secure remain ‘food secure.’ See *Table 1* for a summary of these classifications.

Table 1

Food Security Classification Variables		
Variable	Description	Values
FS_score	Standard classification	1: Food Secure 2: Low Food Security 3: Very Low Food Security
FS_score_binary	Binary classification	0: Food Secure 1: Food Insecure

Figure 1 illustrates the distribution of the FSS data in New York State by classification type. As shown below, the data is unbalanced: 89% of households are food secure while 11% are food insecure. This 11% comprises 6.9% categorized as low food security and 4% as very low food security.

Figure 1



3.2. Covariate Identification for Baseline Models

The baseline model was constructed using common demographic features present in both the ACS and the FSS datasets. These features were selected based on common availability and relevance to food access. All categorical variables were transformed into binary variables. Furthermore, covariates with discrepancies between the ACS and FSS were recoded to ensure uniformity across both datasets. A comprehensive list and description of these covariates can be found in Appendix A, *Table A1*, while summary statistics for both the FSS and ACS datasets are presented in Appendix A, *Table A2*.

3.3. Supplemental Predictor Variable Identification for Extension Models

In the FSS, a household's food security classification is determined by its responses to ten food-related questions. These questions are shown in *Table 2*, below, along with an additional question about WIC. For households without children, a classification of low food security is assigned if the response is affirmative to three or more of these ten questions, while very low food security is assigned for affirmative responses to six or more. The 11 questions in *Table 2* (referred to as Supplemental Predictors) are tested in the extension models.²

² Note that in 2022, the USDA made changes to the FSS. Most of the changes involved a reordering of questions or an update of language. Upon review of these changes, none of the questions analyzed in this report had substantial changes. See Matthew Rabbitt et al., "[Household Food Security in the United States in 2022](#)," 2023, pages 2-3.

Households with children aged 0 to 17 are asked additional questions about food access, and their food security is evaluated based on slightly different criteria.³ To maintain the generalizability of this study, these child-specific questions are not included in the analysis.

Table 2

Supplemental Predictor Variables and Associated Survey Questions			
Variable Name	Question	Affirmative (1)	Negative (0)
food_run_out	“We worried whether our food would run out before we got money to buy more.” Was that often, sometimes, or never true for you in the last 12 months?	Often or Sometimes	Never
food_not_last	“The food that we bought just didn’t last, and we didn’t have money to get more.” Was that often, sometimes, or never true for you in the last 12 months?	Often or Sometimes	Never
no_balanced_meals	“We couldn’t afford to eat balanced meals.” Was that often, sometimes, or never true for you in the last 12 months?	Often or Sometimes	Never
cut_skip_meals	In the last 12 months, did you or other adults in the household ever cut the size of your meals or skip meals because there wasn’t enough money for food?	Yes	No
cut_skip_meals_often	(If yes to <i>cut_skip_meals</i>) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?	Almost every month or some months	Only 1 or 2 months
eat_too_little	In the last 12 months, did you ever eat less than you felt you should because there wasn’t enough money for food?	Yes	No
hungry_but_not_eat	In the last 12 months, were you ever hungry, but didn’t eat, because there wasn’t enough money for food?	Yes	No
lose_weight	In the last 12 months, did you lose weight because there wasn’t enough money for food?	Yes	No
not_eat_whole_day	In the last 12 months, did you or other adults in your household ever not eat for a whole day because there wasn’t enough money for food?	Yes	No
not_eat_whole_day_often	(If yes to <i>not_eat_whole_day</i>) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?	Almost every month or some months	Only 1 or 2 months
WIC	During the past 30 days, did any (women/women or children/children) in this household get food through the WIC program?	Yes	No
Notes and Sources: Questions and food security classification from "Household Food Security in the United States in 2022," US Department of Agriculture Economic Research Service, October 2023. Variable names as represented in associated charts.			

³ For more information, see Matthew Rabbitt et al., “[Household Food Security in the United States in 2022](#),” 2023, page 5.

3.4. Identification of the Baseline Models

This analysis established two baseline random forest classification models: one using the standard three-class food security classification (Standard Baseline Model) and the other using the binary classification (Binary Baseline Model). These models were trained on FSS data that had been cleaned to reflect only the covariates shared with the ACS data.

To optimize model performance for each classification type, the analysis tested different covariates and cycled through two hyperparameters: class weight and tree depth. Class weights were assigned to different food security classes to address the inherent data imbalance, where food-secure households significantly outnumbered food-insecure ones. Specifically, higher weights were applied to the food-insecure classes to encourage the model to prioritize their correct identification and to impose a greater penalty for their misclassification. For each tested class weight, the optimal tree depth was determined based on the minimum validation sample loss.

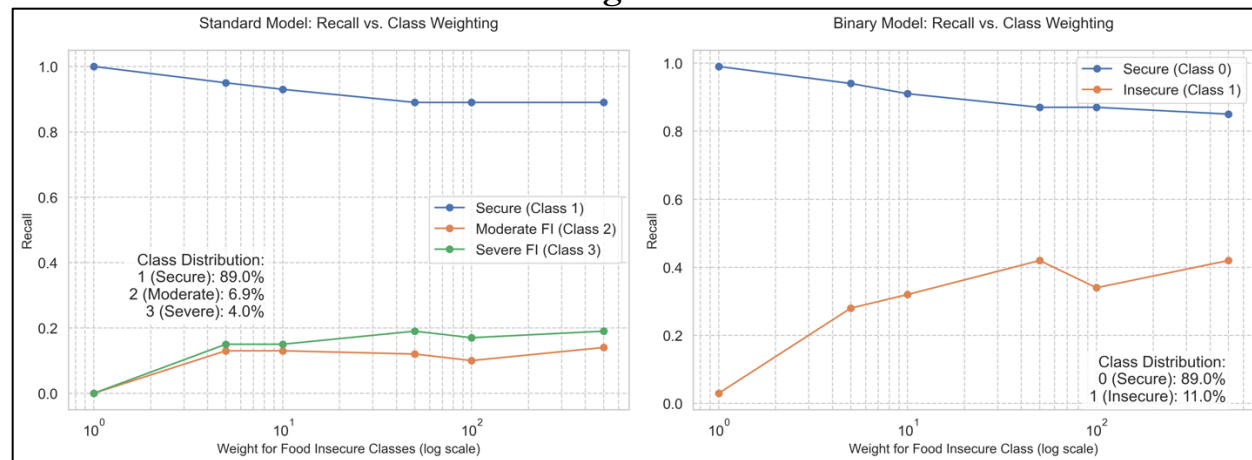
Finally, the optimal class weight was selected based on the maximum recall achieved for the food insecure class. This selection criterion aimed to maximize the proportion of actual food-insecure households that the model correctly identified. As shown in *Figure 2*, increasing the weights of the food insecure classes led to an improvement in their recall.

The recall of the food secure class and the area under the curve (AUC) were also considered in the model optimization process. Although the recall for the food-secure class experienced a slight decrease as food insecure weights increased, the drop was not substantial (see *Figure 2*). For both baseline models, the AUC remained between 0.7 and 0.8.

For both the Standard and Binary Baseline Models, the food-insecure classes were assigned an optimal weight 50 times greater than that of the food-secure class. The optimal tree depth determined through the optimization process was 15 for the Standard Baseline Model and 16 for the Binary Baseline Model.

The Standard Baseline Model and Binary Baseline Model identified in this process are evaluated on their predictive power in the Results section.

Figure 2



3.5. Model Extension

To assess the predictive strength of the individual FSS survey questions (Supplemental Predictors), which are not included in the ACS data, these variables were incorporated into the Standard and Binary Baseline Models. This analysis employed two distinct versions of the model extension.

3.5.1. Version 1: Retrain Model Using Predictor Features:

In Version 1, for each Supplemental Predictor, a new random forest model was retrained. This retraining used the same optimized hyperparameters (tree depth and class weighting) and the same set of baseline covariates identified earlier. Each new model incorporated either one or two additional Supplemental Predictors at a time, while all other inputs and hyperparameters remained constant. Model performance was evaluated by the associated increase in recall for the food insecure classes compared to the baseline recall. An improvement in recall indicates that the added predictor enhanced the model's ability to correctly identify food-insecure households that the baseline model had misclassified.

3.5.2. Version 2: Prediction Improvement:

In Version 2, the original, pre-trained baseline models remain unchanged. Rather than retraining the models, this approach evaluates whether each Supplemental Predictor could have helped correct misclassifications made by the baseline models.

For each Supplemental Predictor:

- 1) The model's predictions were generated using the original baseline.
- 2) Cases that were misclassified (food insecure households predicted as food secure) were examined to see whether the Supplemental Predictor could have helped distinguish them.
- 3) Improvement was measured by assessing whether the inclusion of the predictor could reduce false negatives, or the missed food insecure cases.

This approach simulates the impact of adding a single FSS question to an existing survey like the ACS without re-estimating the model, providing insight into marginal gains in predictive power.

4. Results

4.1. Baseline Model

As detailed below, the baseline models are poor predictors of food insecurity. Thus, a model trained solely on FSS data, using only the covariates shared with the ACS, would not accurately predict the food insecurity status of households within the ACS dataset in its current form.

4.1.1. Standard Baseline Model

Figure 3 shows the confusion matrix of the Standard Baseline Model. This figure indicates that the model predominantly classified all observations as food secure. Only 12% of moderately food insecure

and 19% of severely food insecure households were correctly classified. A confusion matrix displaying the absolute counts of model errors is available in *Figure A1* of Appendix A.

Figure 3

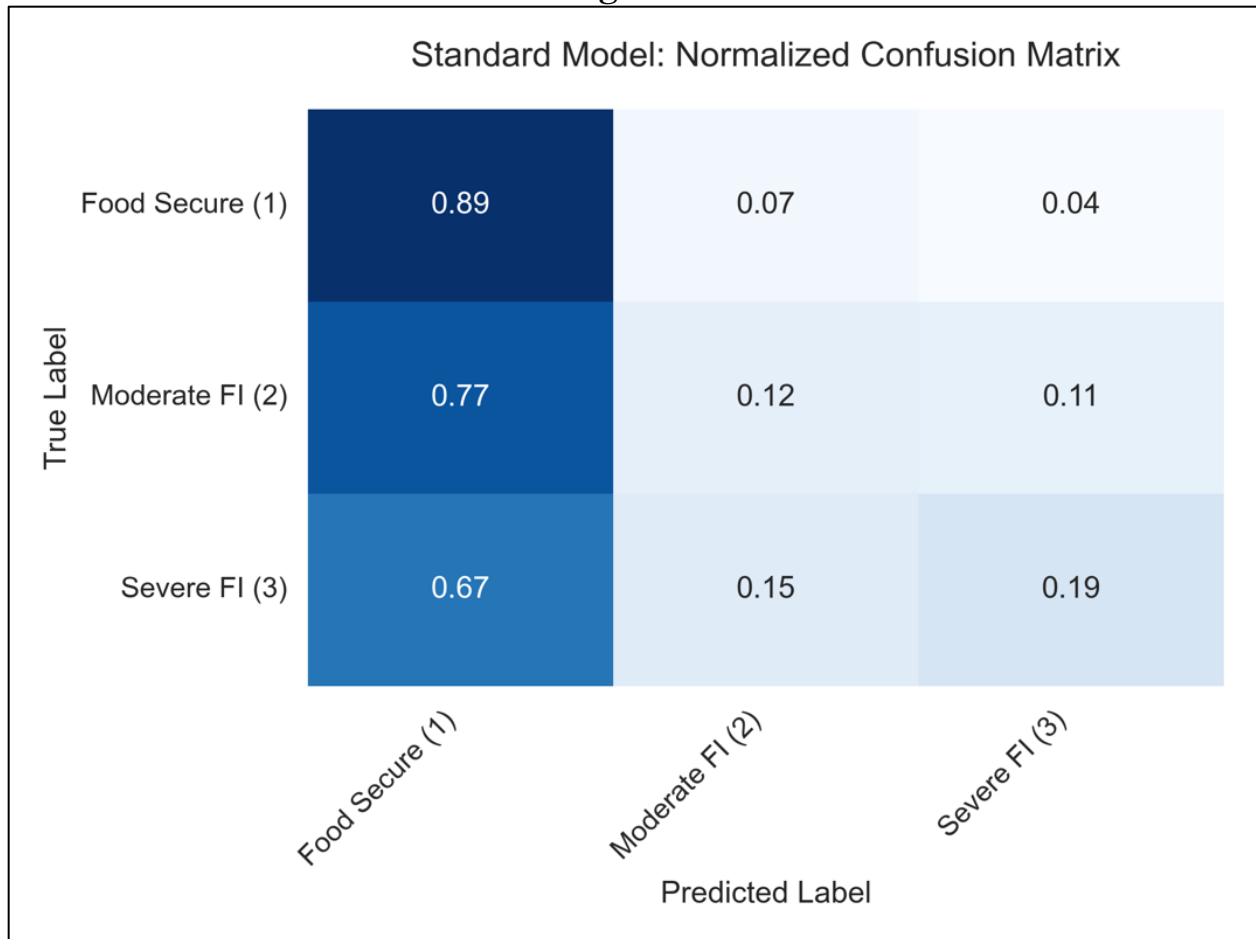
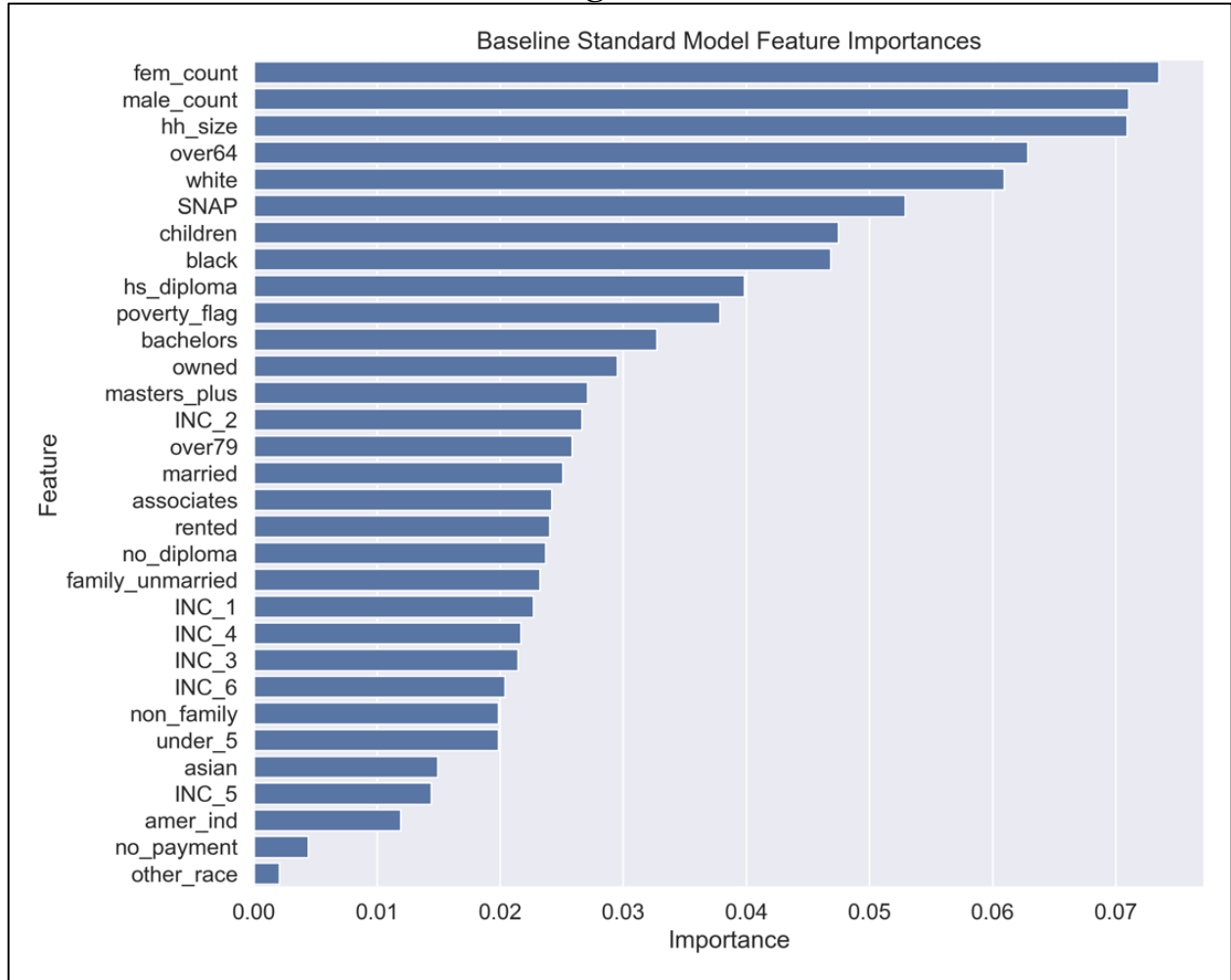


Figure 4 displays the feature importance scores of each covariate. All covariates exhibit low importance (below than 0.1); this is another indication of a weak model. Surprisingly, most income-related indicators rank low on feature importance.

Overall, the model correctly classifies a high percentage of households. The AUC was 0.75, suggesting moderate performance. Yet, this high accuracy is largely attributable to the substantial proportion of food-secure households in the dataset, and the model failed to correctly identify a significant number of food-insecure households.

Figure 4



4.1.2. Binary Baseline Model

Figure 5 presents the confusion matrix of the Binary Baseline Model. As with the Standard Baseline Model, the majority of all observations were classified as food secure. However, this model demonstrates improved performance compared to the Standard Baseline Model, correctly identifying 39% of food-insecure households. The confusion matrix displaying the absolute counts of model errors is available in *Figure A2* of Appendix A.

Figure 5

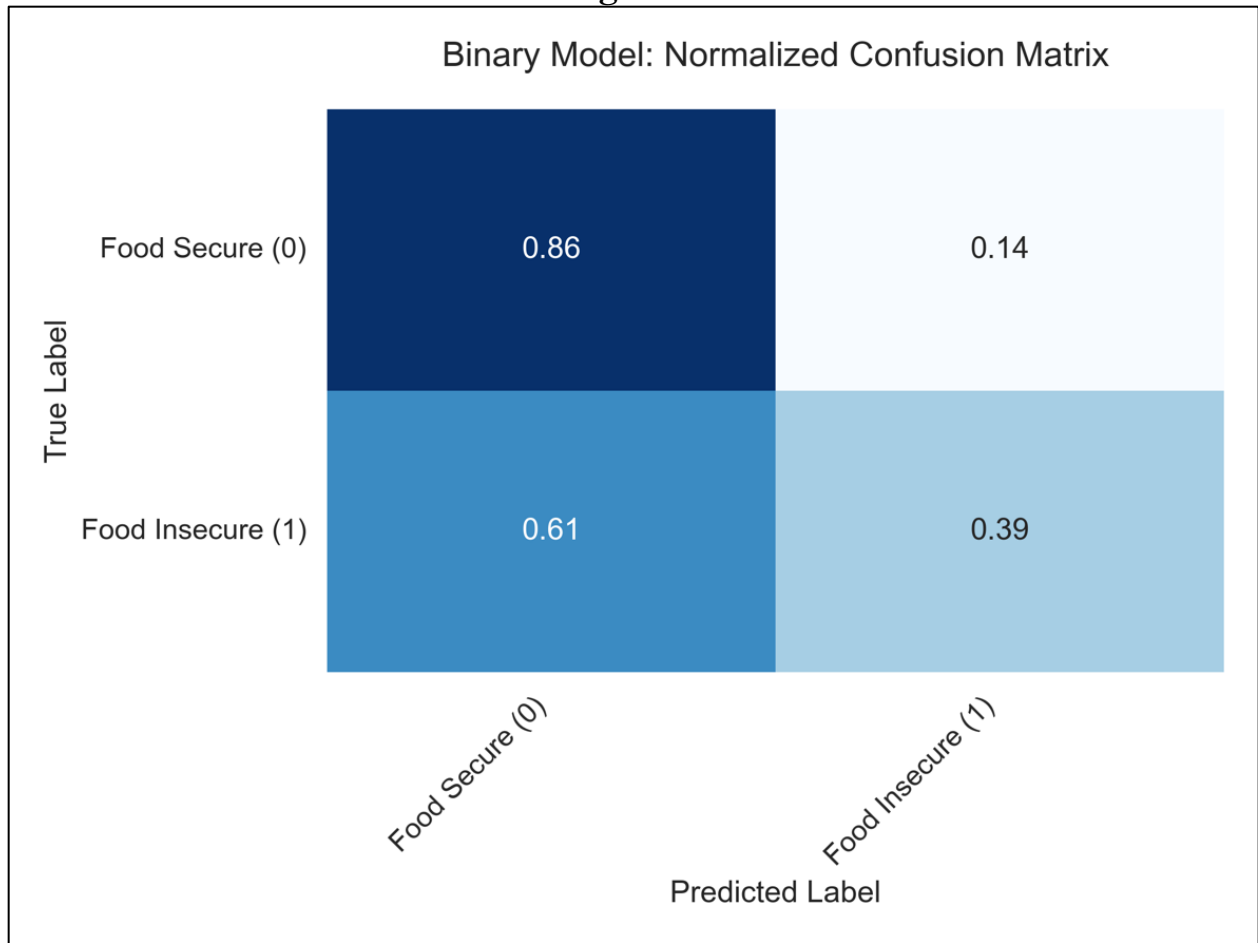
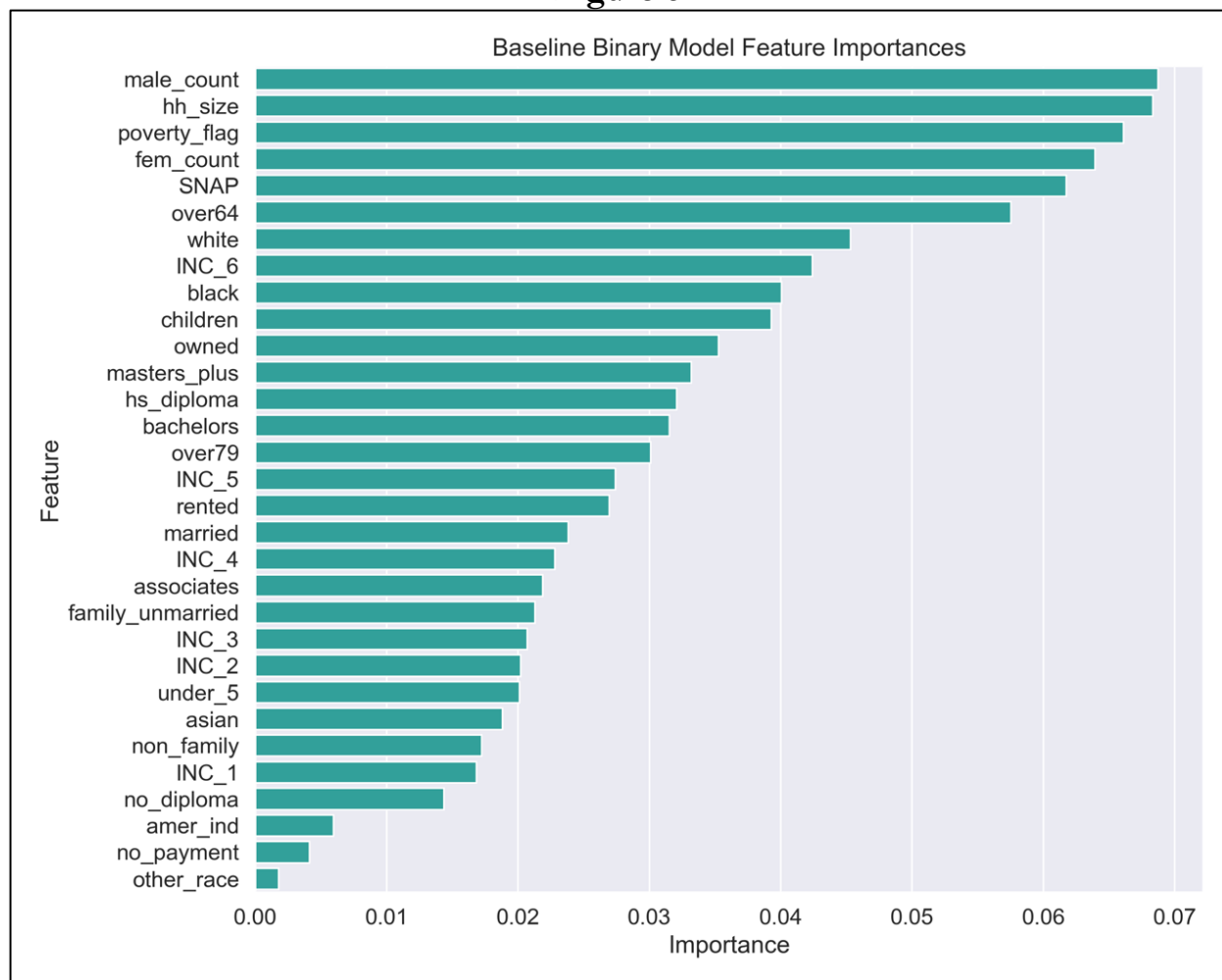


Figure 6 shows the feature importance scores of each covariate. All covariates have low importance (below 0.1), again indicating a weak model. The income indicators seem to hold relatively higher importance than those in the Standard Baseline Model, but they are still low.

As with the Standard Baseline Model, most households are correctly identified because the majority of households are food secure. The AUC is 0.75, further suggesting moderate performance overall. However, the model poorly predicts food insecurity.

Figure 6



4.2. Model Extension: Version 1

4.2.1. Standard Version 1

The incorporation of the Supplementary Predictors dramatically increases the recall of the food insecure classes. *Figure 7* shows average recall improvement for the moderate food insecure (class 2) and severely food insecure (class 3) households. When these predictors were added to the model collectively, the two questions yielding the most significant improvement in recall (a 0.7 increase) were:

- 1) *food_not_last*: “The food that we bought just didn’t last, and we didn’t have money to get more.” Was that often, sometimes, or never true for you in the last 12 months?
- 2) *cut_skip_meals_often*: How often did [you or other adults in the household ever cut the size of your meals or skip meals because there wasn’t enough money for food?]*—almost every month, some months but not every month, or in only 1 or 2 months?*

Figure 7

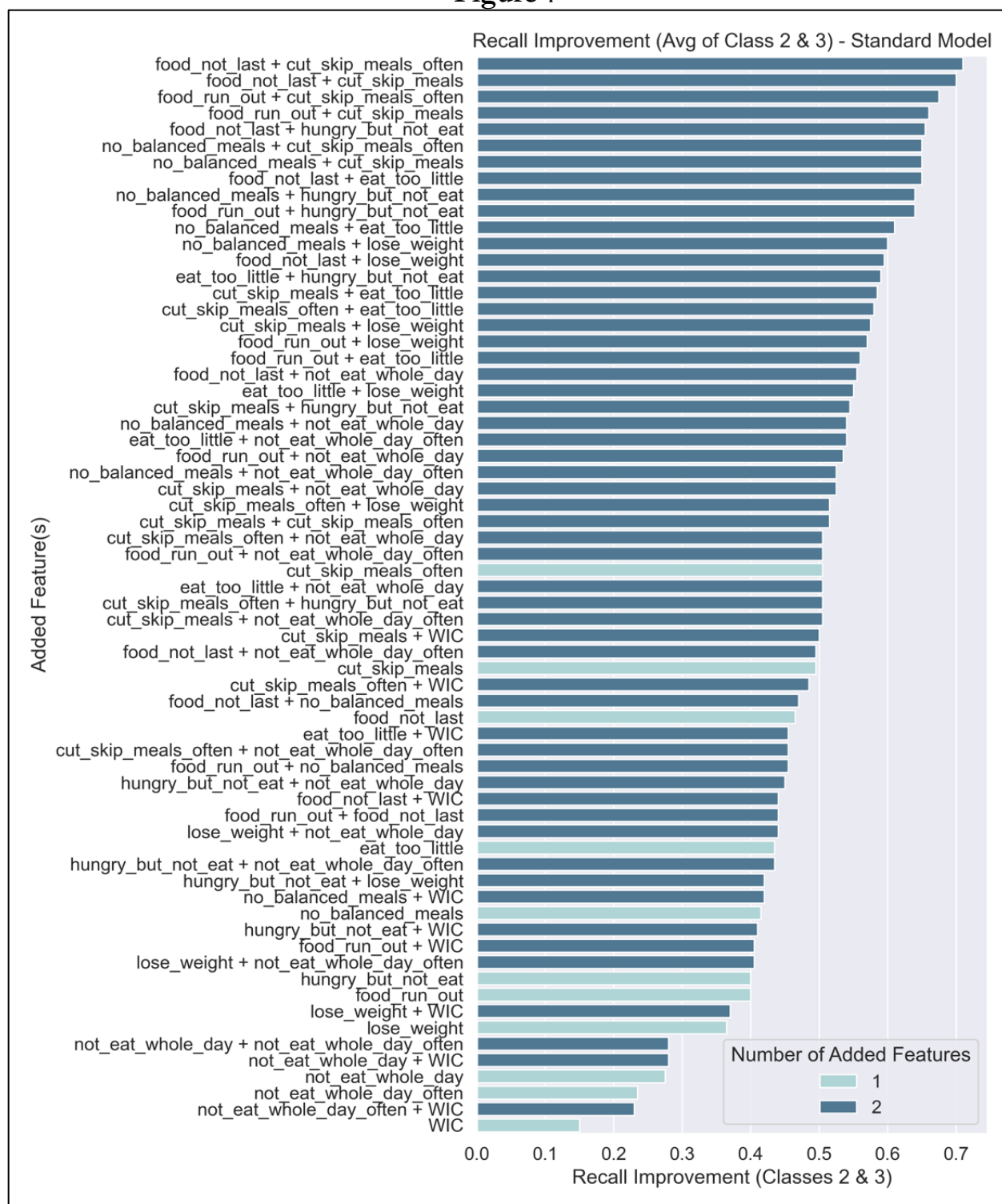
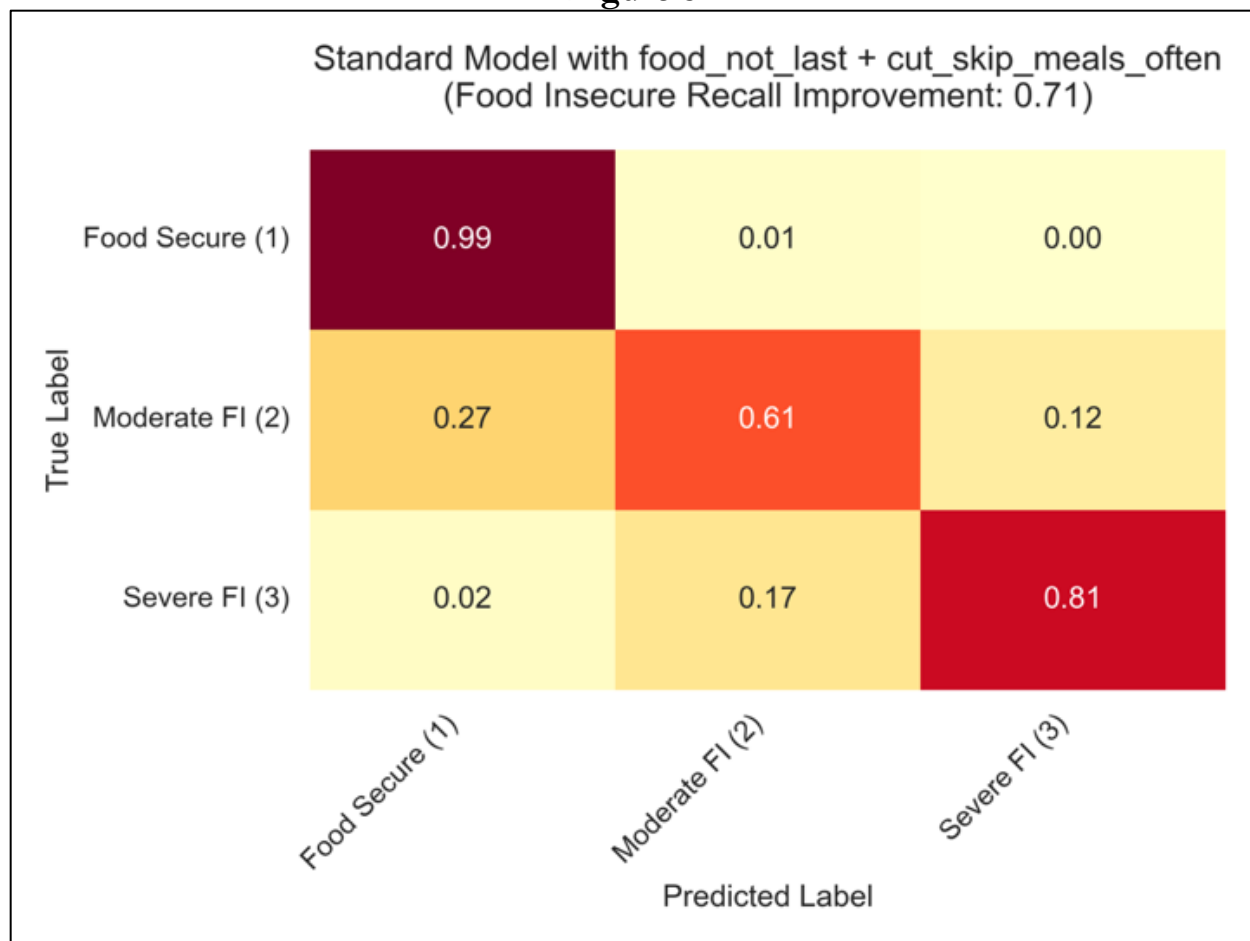


Figure 8 shows the confusion matrix for the model incorporating these two Supplemental Predictors. The recall for the food-secure class reached 0.99, and for the severely food-insecure class reached 0.81. However, the recall for the moderately food-insecure class remained considerably lower at 0.61.

The variable *cut_skip_meals_often* was also the single question yielding the strongest recall improvement, at roughly 0.5 (see *Figure 7*). Surprisingly, the variable *WIC*, which indicates whether the household has any WIC recipients, does not substantially improve recall.

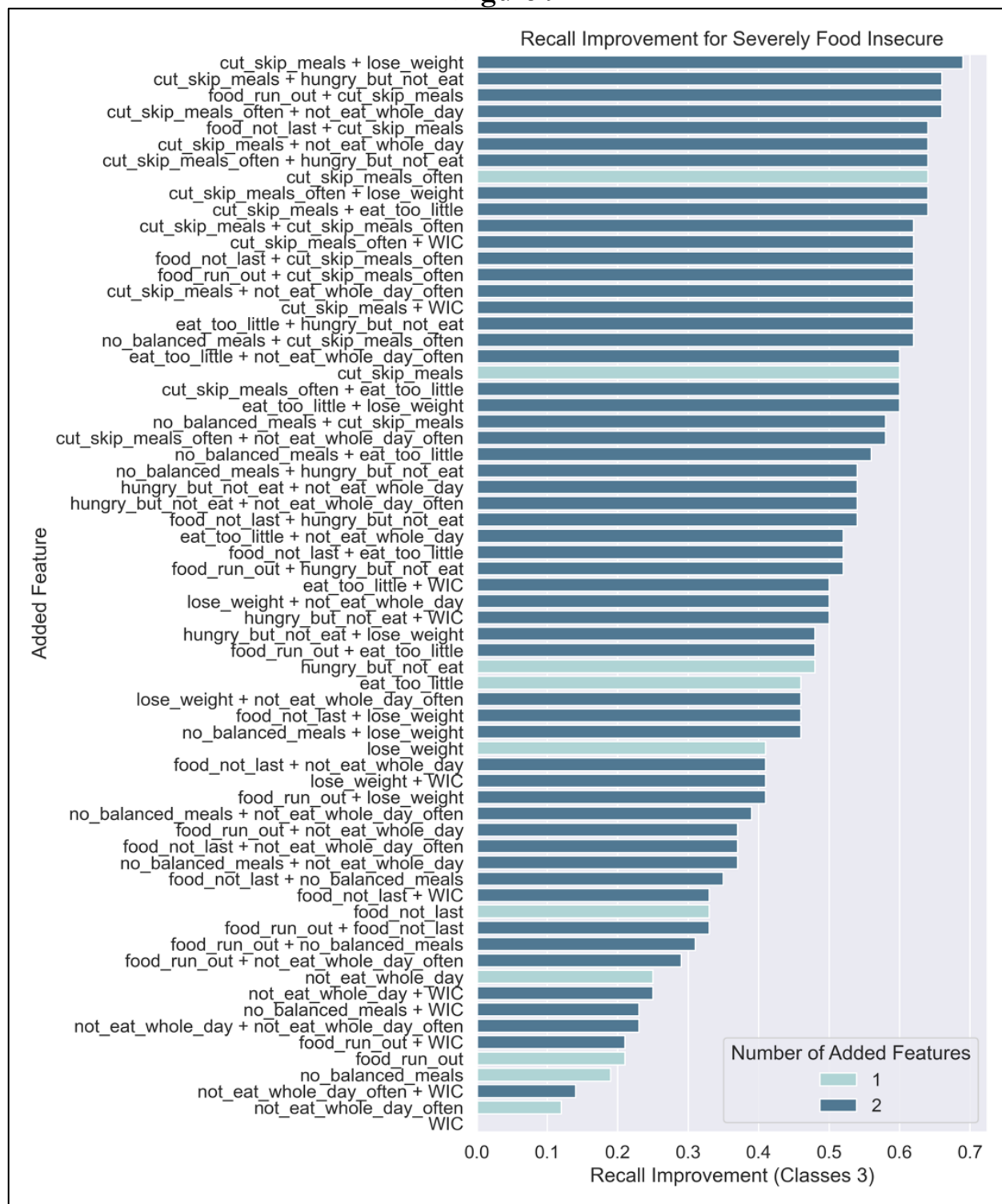
Figure 8



In contrast, the Supplemental Predictors that, when added to the model in combination, yielded the most significant improvement in recall specifically for the severely food-insecure class (as depicted in *Figure 9*) were:

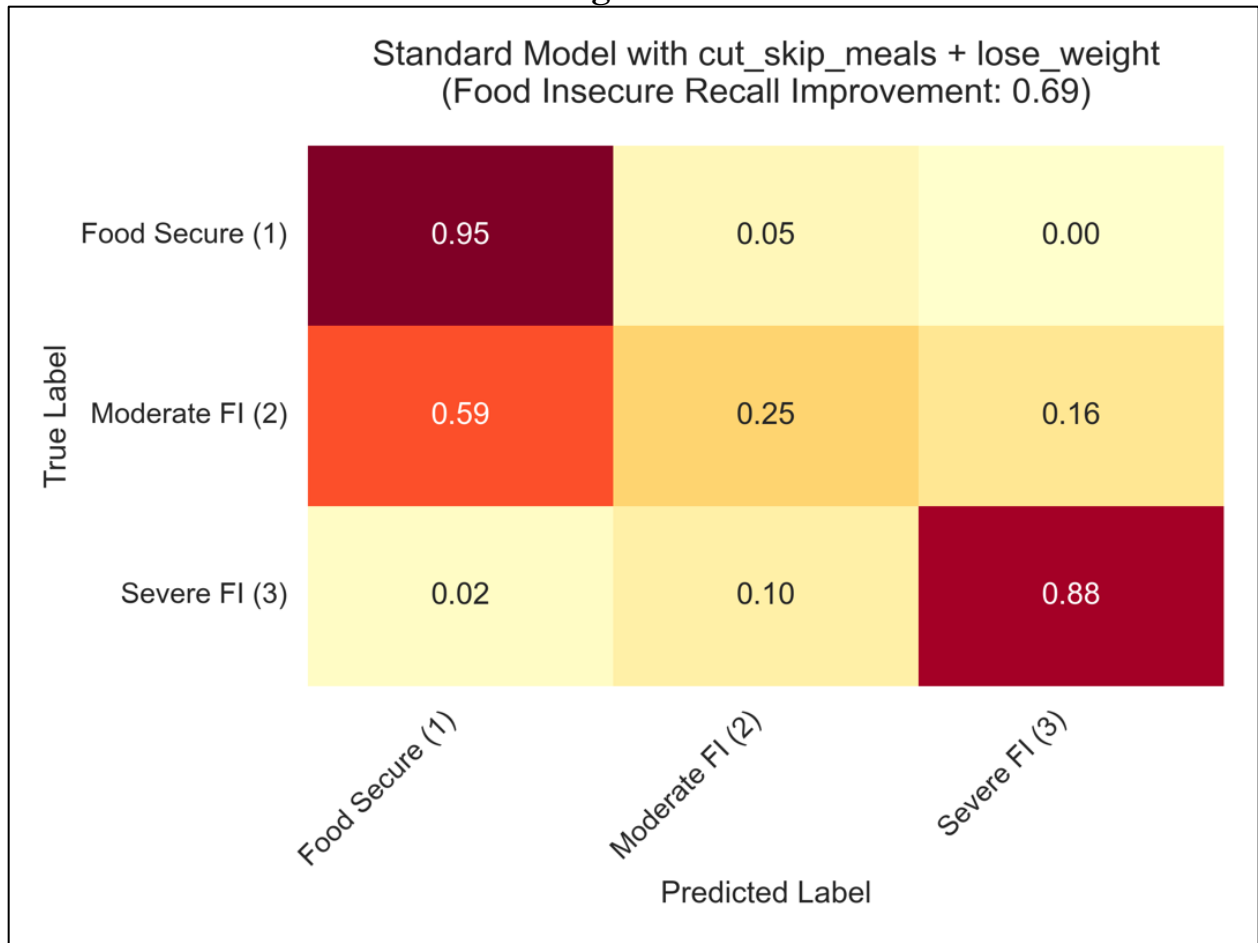
- 1) *cut_skip_meals*: In the last 12 months, did you or other adults in the household ever cut the size of your meals or skip meals because there wasn't enough money for food?
- 2) *lose_weight*: In the last 12 months, did you lose weight because there wasn't enough money for food?

Figure 9



The confusion matrix for this model is shown in *Figure 10*. The model correctly predicts 88% of severely food insecure households and 95% of food secure households, but only 25% of moderately food insecure households. Notably, most moderately food insecure households were misclassified as food secure, as opposed to severely food insecure.

Figure 10



4.2.2. Binary Version 1

As with the standard model, incorporating the Supplemental Predictors also enhanced the recall for the food-insecure class in the binary model. *Figure 11* illustrates the average recall improvement for the food insecure households. When added to the model together, the following questions bring the largest recall improvement (a 0.5 increase):

- 1) *no_balanced_meals*: “We couldn’t afford to eat balanced meals.” Was that often, sometimes, or never true for you in the last 12 months?
- 2) *eat_too_little*: In the last 12 months, did you ever eat less than you felt you should because there wasn’t enough money for food?

Figure 11

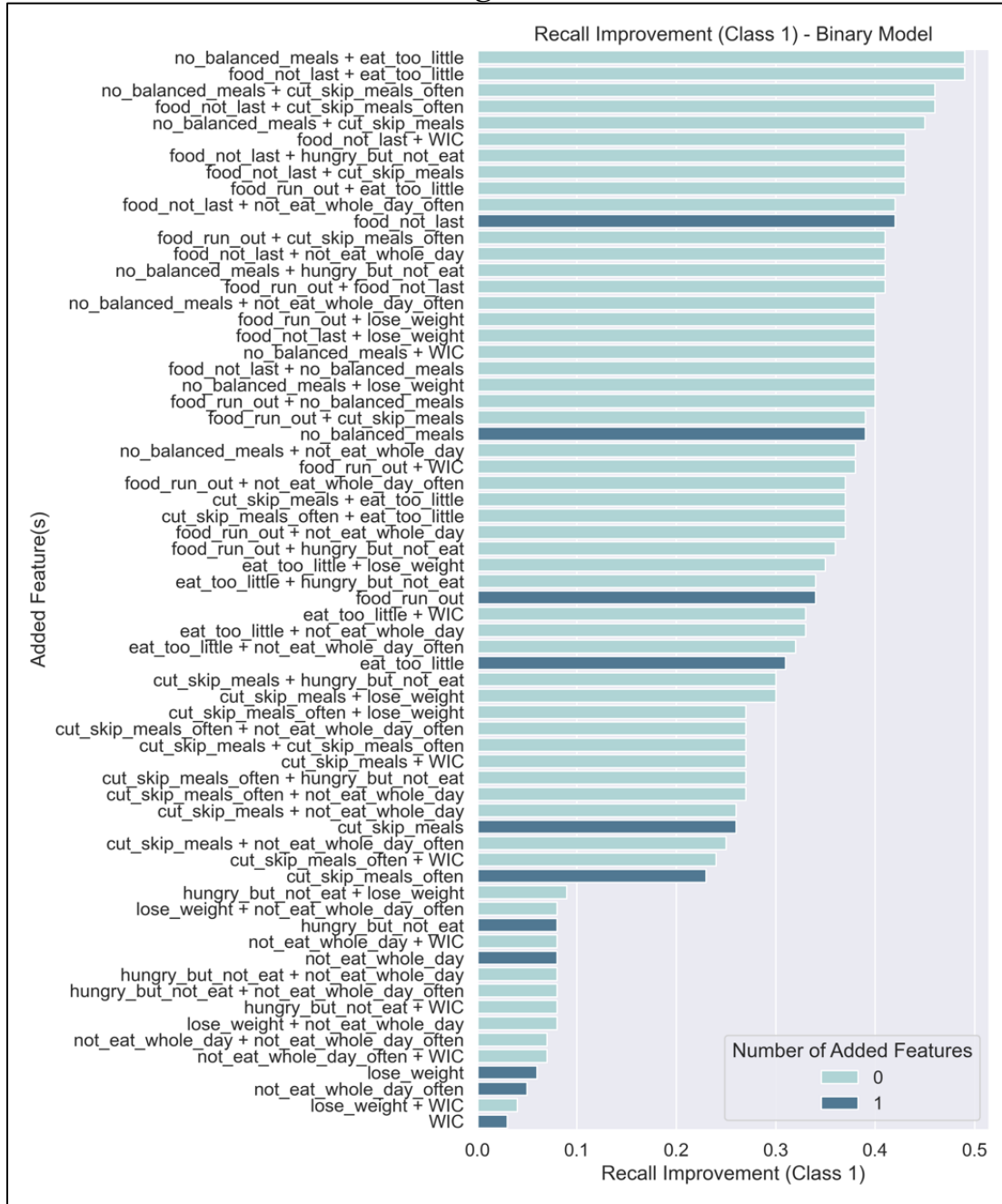
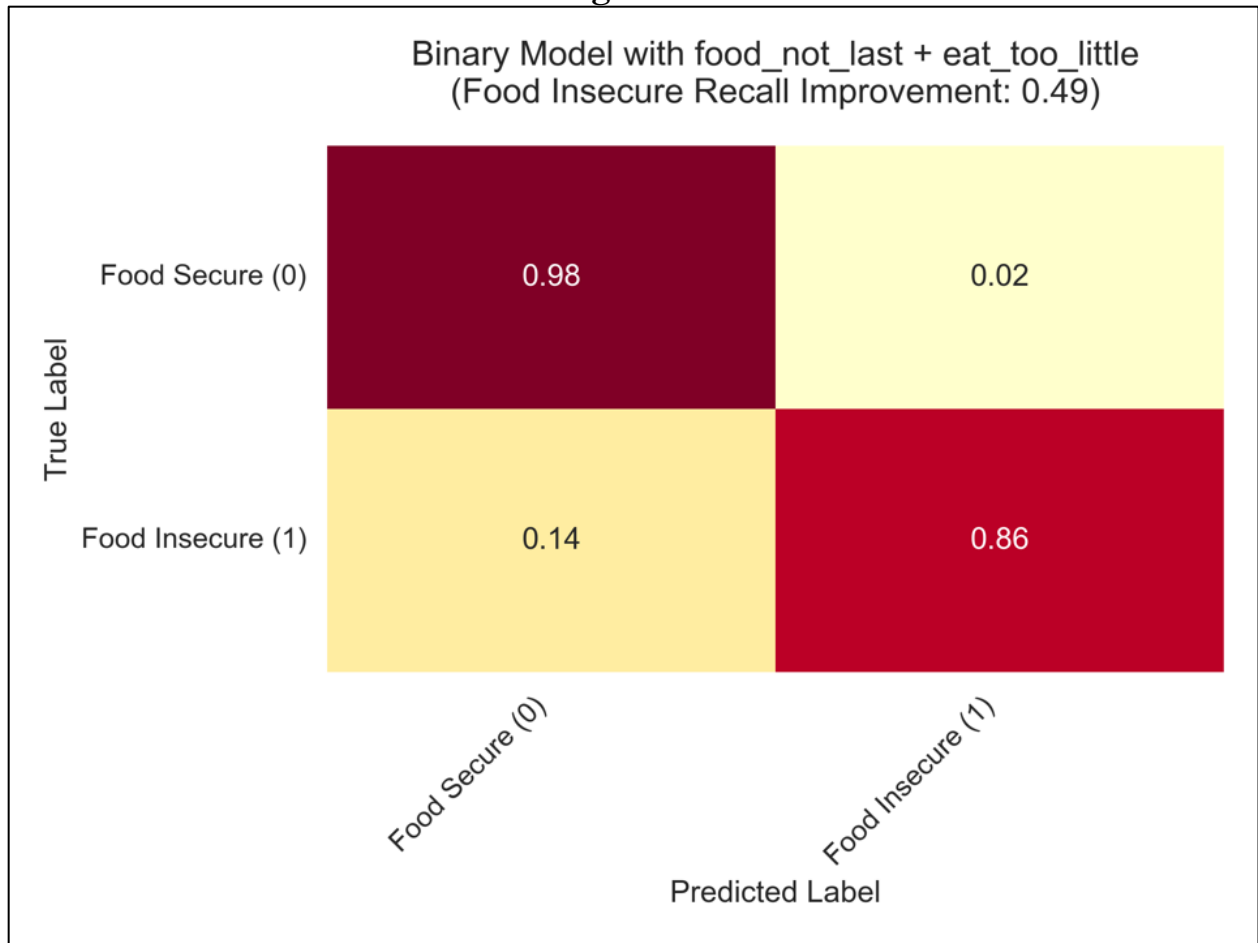


Figure 12 shows the confusion matrix for the model that includes these two variables. The food secure recall is 0.98 and the food insecure recall is 0.86, a significant improvement upon the Binary Baseline Model.

The variable *food_not_last* was the single question that, when added individually, yielded the strongest recall improvement, approximately 0.42 (see Figure 7). As seen with the standard model, the variable *WIC*, did not substantially improve recall.

Figure 12

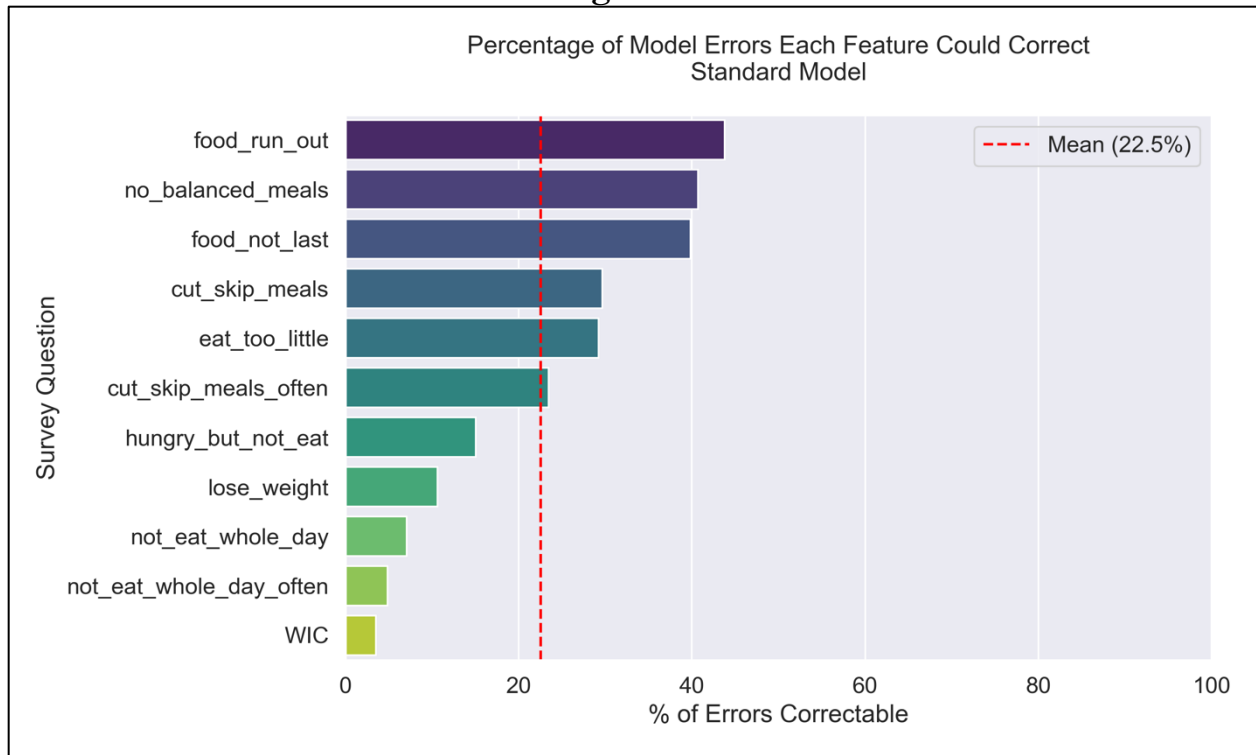


4.3. Model Extension: Version 2

4.3.1. Standard Version 2

The results of model extension Version 2 complement the findings from Version 1. As summarized in *Figure 13*, on average, 22.5% of the Standard Baseline Model's false negative predictions for food insecurity could potentially be corrected by adding a single Supplementary Predictor. The variable *food_run_out* ("We worried whether our food would run out before we got money to buy more") showed the greatest potential to improve recall (44%), followed by *no_balanced_meals* and *food_not_last* (both close to 40%). In other words, *food_run_out* was affirmative in 44% of false negatives among the food insecure classes; by including *food_run_out* in the model, it might have helped correctly identify those households. This is supported by the finding that 87.5% and 97% of moderately and severely food insecure households answered affirmatively to *food_run_out* (See Appendix A, *Figure A3*).

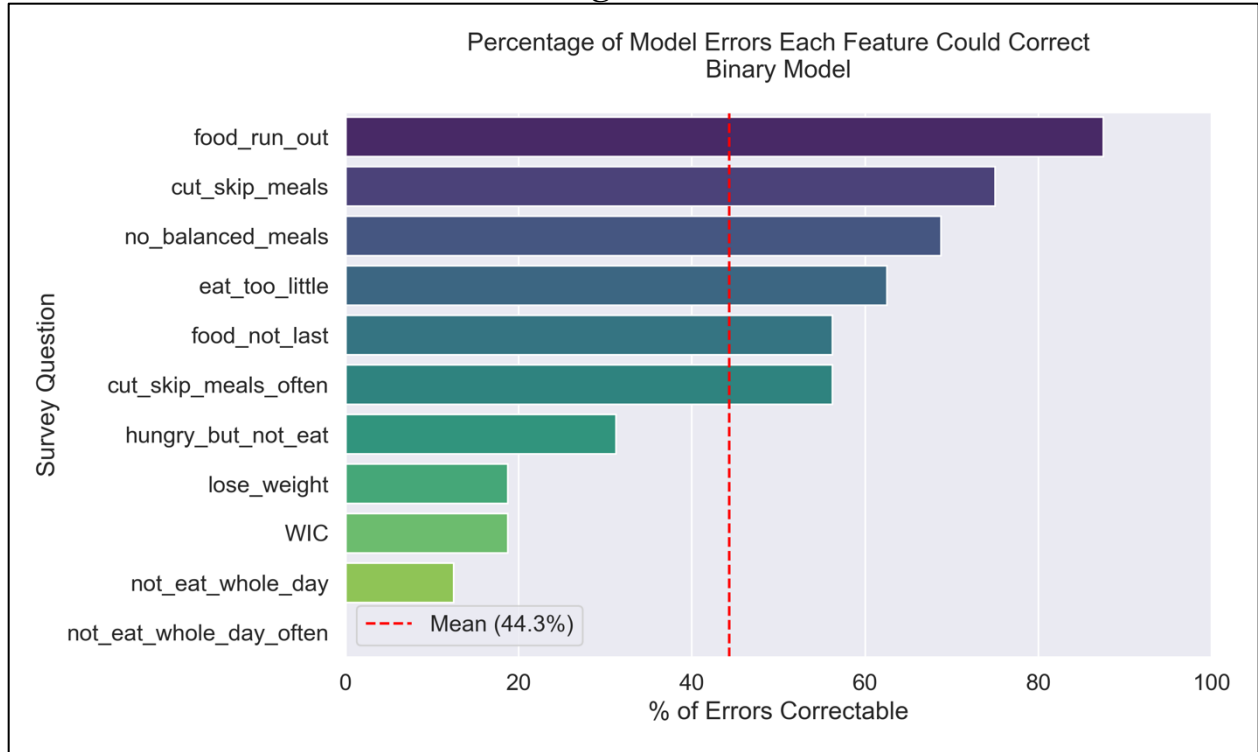
Figure 13



4.3.2. Binary Version 2

Figure 14 shows the results of the binary Version 2 extension. As with the standard extension, *food_run_out* shows the greatest potential for improving recall. Specifically, 87% of the model's false negatives (households that were actually food insecure but predicted as food secure) had answered affirmatively to *food_run_out*. This suggests that including this feature in the model could potentially help correctly identify a large share of misclassified food insecure households. This is supported by the finding that 91% of food insecure households answered affirmatively to *food_run_out*.

Figure 14



5. Discussion

The findings presented in the prior section lead to two primary conclusions: 1) a random forest classification model trained solely on the covariates common to both the FSS and the ACS exhibits limited accuracy in predicting food insecurity; and 2) the addition of just one or two food-security variables dramatically enhances the model's predictive power.

A random forest classification model cannot be trained on existing FSS survey data to accurately predict food insecurity for households in the unlabeled ACS data. As shown in *Figures 3* and *5*, only 12% to 40% of food insecure households were correctly identified by these baseline models. The unbalanced data is likely a large contributing factor to this poor model performance, as only 11% of households in the FSS are food insecure.

However, the model's predictive power improves considerably with the inclusion of just one or two food security-related variables. The following variables and corresponding questions demonstrated the most significant impact on predicting food insecurity:

- 1) *cut_skip_meals*: In the last 12 months, did you or other adults in the household ever cut the size of your meals or skip meals because there wasn't enough money for food?
- 2) *lose_weight*: In the last 12 months, did you lose weight because there wasn't enough money for food?
- 3) *food_not_last*: "The food that we bought just didn't last, and we didn't have money to get more." Was that often, sometimes, or never true for you in the last 12 months?

- 4) *no_balanced_meals*: “We couldn’t afford to eat balanced meals.” Was that often, sometimes, or never true for you in the last 12 months?

With the addition of these variables or variable combinations, up to 88% of food insecure households can be correctly classified. By contrast, other variables such as *not_eat_whole_day* (In the last 12 months, did you or other adults in your household ever not eat for a whole day because there wasn’t enough money for food?) and WIC provide little to no increase in predictive power. This is likely because relatively few households responded affirmatively to these questions (see *Figure A3* in Appendix A).

Overall, the results from this analysis suggest that the incorporation of just one or two food security-related questions into the ACS could enable a machine learning model to identify food-insecure households with relatively high accuracy. Given the broad coverage of the ACS, this information could play a vital role in helping policymakers identify areas at high risk of food insecurity and make targeted interventions.

It is important to note that this analysis was conducted using household data specifically from New York State; consequently, the generalizability of these estimates is limited to this region. An extension of this analysis can utilize data from the entire U.S. and investigate whether the results differ by state. Furthermore, a significant limitation pertains to the definition of food security employed in the FSS, which primarily focuses on undernutrition, or the lack of sufficient food. The survey includes fewer and less direct questions about food quality and overnutrition, or the overconsumption of unhealthy food. Overnutrition is a significant health and economic burden in New York and the U.S. yet is largely absent in this measure of food security. Therefore, the findings of this analysis primarily pertain to undernutrition and offer limited insights into the overnutrition dimension of food insecurity.

In conclusion, food insecurity remains a persistent issue in New York and the U.S. Through machine learning, researchers and policymakers can potentially leverage and improve upon existing resources to more effectively identify at-risk populations and develop targeted, efficient policies aimed at reducing food insecurity.

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7. Appendices

7.1. Appendix A

Table A1

Covariates Used for Baseline Model Common Across ACS and FSS Surveys	
Variable	Description
<i>Continuous Variables</i>	
hh_size	Number of persons in household
children	Number of children
under_5	Number of residents under 5 years old
over64	Number of residents over 64 years old
over79	Number of residents over 79 years old
fem_count	Number of female residents
male_count	Number of male residents
<i>Binary Variables</i>	
hs_diploma	Highest level of education - High School
no_diploma	Highest level of education - Less than High School
associates	Highest level of education - Associate's
bachelors	Highest level of education - Bachelor's
masters_plus	Highest level of education - Master's or higher
owned	Household is owned
rented	Household is rented
no_payment	Household is occupied without payment
SNAP	Household received SNAP
married	Married household
family_unmarried	Family household unmarried
non_family	Non-family household
white	Reference person race - White
black	Reference person race - Black
amer_ind	Reference person race - American Indian
asian	Reference person race - Asian
other_race	Reference person race - Other
poverty_flag	Household is below federal poverty threshold
INC_1	Household Income: \$0 to \$19,999
INC_2	Household Income: \$20,000 to \$39,999
INC_3	Household Income: \$40,000 to \$59,999
INC_4	Household Income: \$60,000 to \$99,999
INC_5	Household Income: \$100,000 to \$149,999
INC_6	Household Income: \$150,000 or more
Note: <i>poverty_flag</i> was calculated using the household size and income, and thresholds identified in: Barbara Guinn and Valerie Figueroa, "General Information System (GIS) Message," June 2023, https://otda.ny.gov/policy/gis/2024/24DC004.pdf .	

Table A2

Covariate Summary Statistics Across FSS and ACS

Variable	Average		Difference
	FSS	ACS	
hh_size	2.34	2.41	-0.07
owned	0.56	0.66	-0.11
rented	0.43	0.32	0.11
no_payment	0.01	0.02	0.00
children	0.46	0.44	0.01
under_5	0.12	0.11	0.01
over64	0.50	0.56	-0.05
over79	0.12	0.12	-0.01
fem_count	1.24	1.25	-0.02
male_count	1.11	1.16	-0.05
hs_diploma	0.31	0.31	0.00
no_diploma	0.04	0.04	0.00
associates	0.10	0.11	-0.01
bachelors	0.27	0.26	0.01
masters_plus	0.26	0.28	-0.01
SNAP	0.10	0.13	-0.04
married	0.42	0.47	-0.05
family_unmarried	0.18	0.16	0.02
non_family	0.40	0.37	0.03
white	0.59	0.76	-0.17
black	0.12	0.11	0.01
amer_ind	0.01	0.02	-0.01
asian	0.07	0.09	-0.02
other_race	0.00	0.09	-0.09
poverty_flag	0.33	0.25	0.08
INC_1	0.15	0.12	0.03
INC_2	0.17	0.13	0.05
INC_3	0.15	0.12	0.03
INC_4	0.20	0.21	-0.01
INC_5	0.14	0.17	-0.03
INC_6	0.18	0.25	-0.07

Figure A1

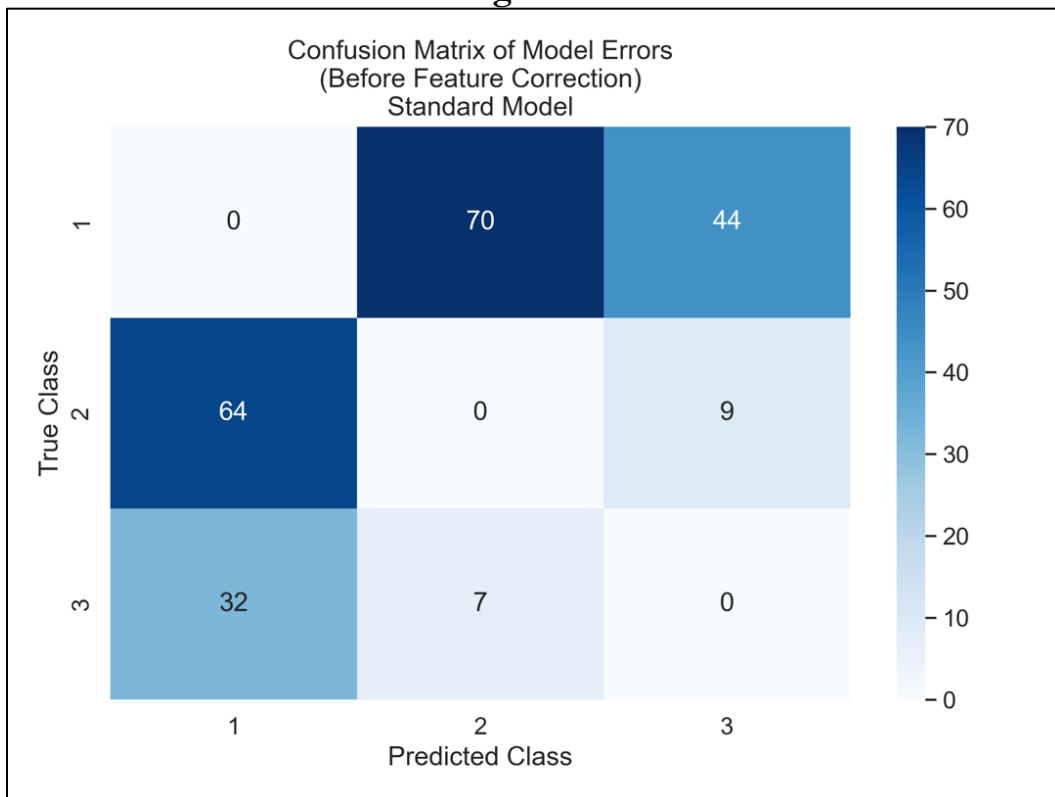


Figure A2

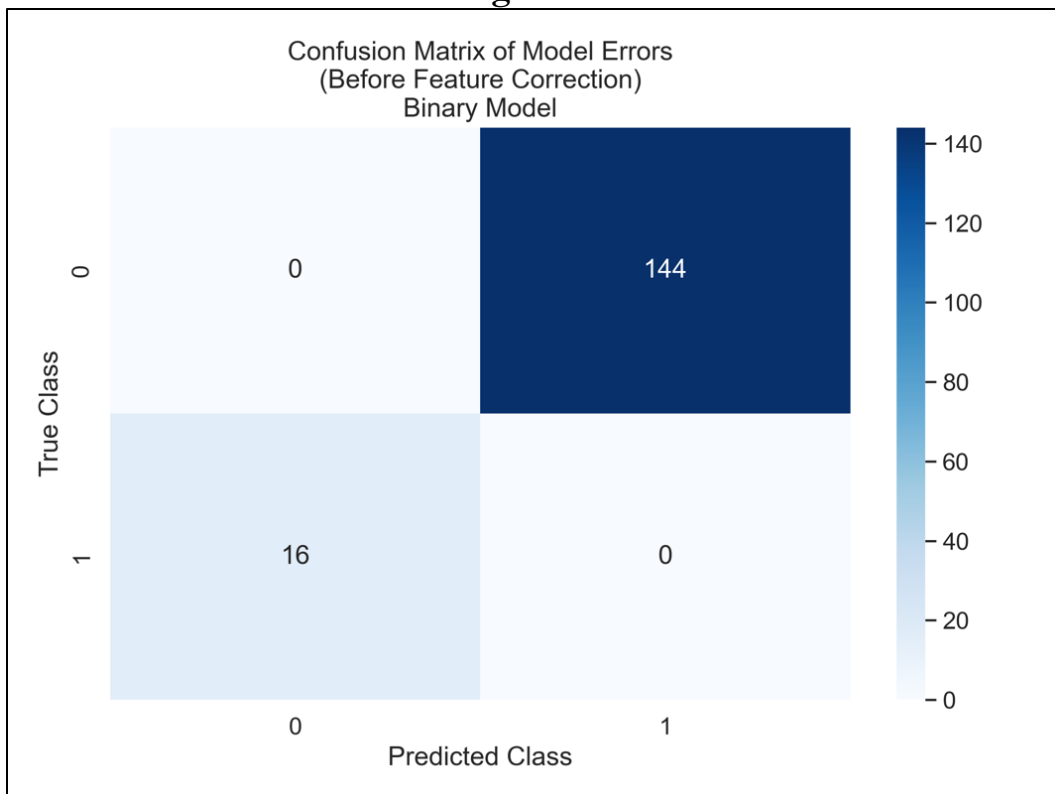


Figure A3

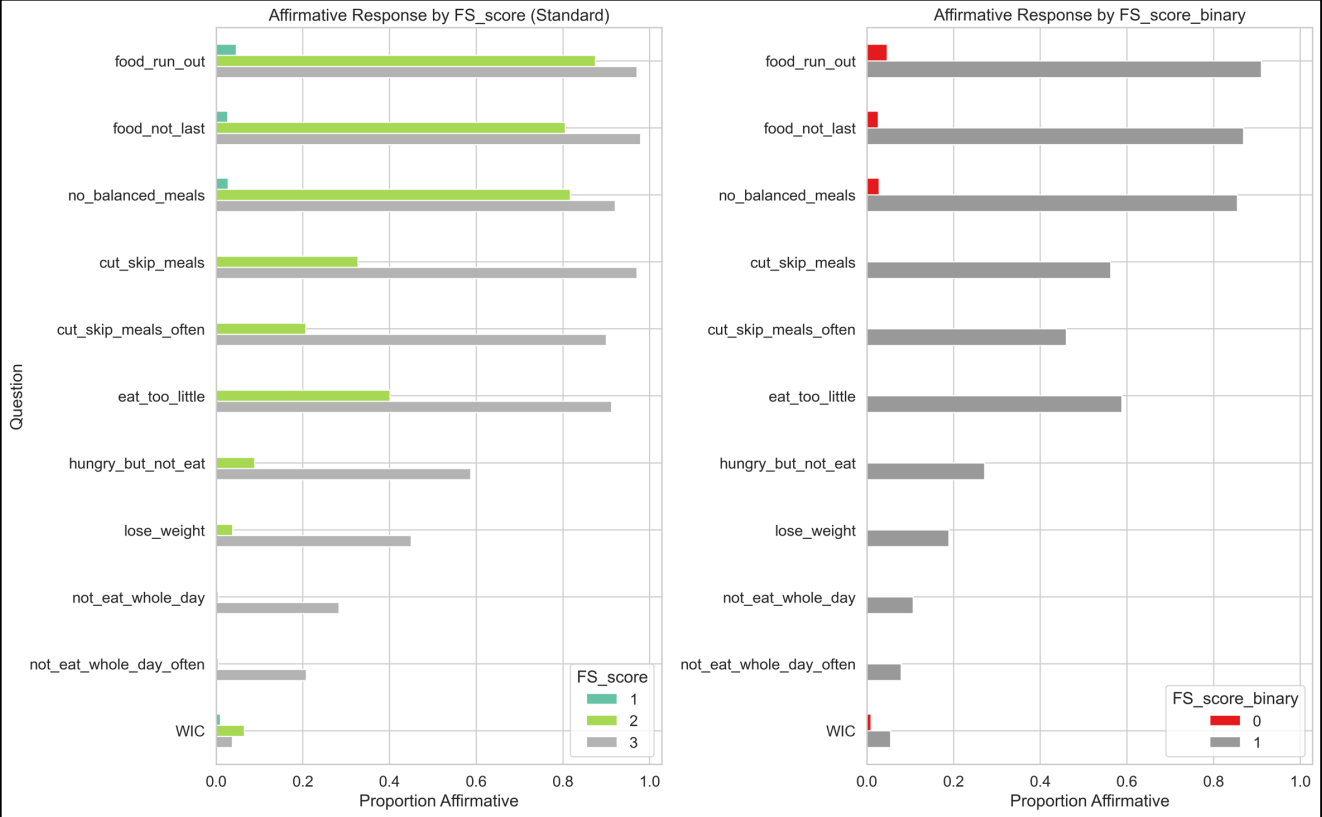


Figure A4

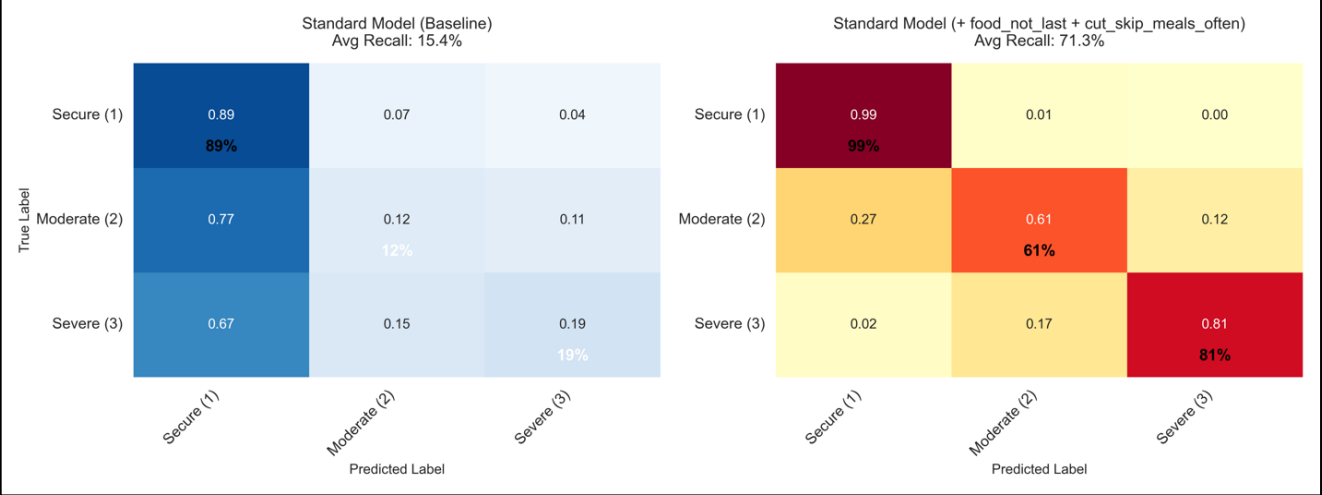
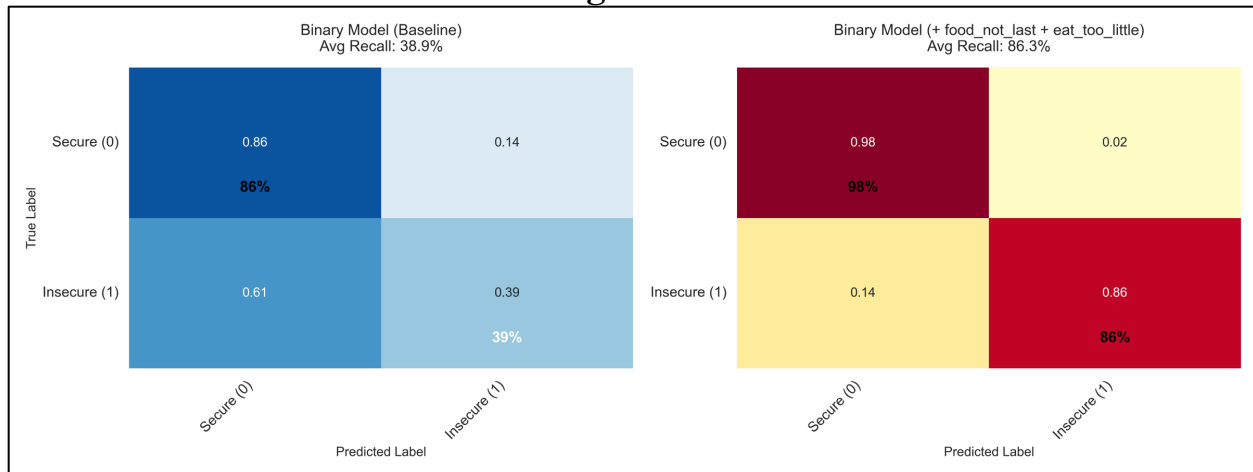


Figure A5



7.2. Appendix B: Data Sources

7.3. Appendix C: Code

ⁱ United States Department of Agriculture, “Food Security in the U.S. - Key Statistics & Graphics,” USDA.gov, January 8, 2025, <https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/key-statistics-graphics>.

ⁱⁱ “NYC Food by the Numbers” (NYC Food Policy, 2024), <https://www.nyc.gov/assets/foodpolicy/downloads/pdf/nycfoodbythenumbersfinal.pdf>.

ⁱⁱⁱ Feeding America, “Understanding Hunger and Food Insecurity,” 2025, <https://www.feedingamerica.org/hunger-in-america/food-insecurity>.

^{iv} Feeding America, “Understanding Hunger and Food Insecurity,” 2025, <https://www.feedingamerica.org/hunger-in-america/food-insecurity>.

^v Jordan Jones, “Supplemental Nutrition Assistance Program (SNAP) - Key Statistics and Research,” www.ers.usda.gov, January 6, 2025, <https://www.ers.usda.gov/topics/food-nutrition-assistance/supplemental-nutrition-assistance-program-snap/key-statistics-and-research>.

^{vi} Jordan Jones, “Supplemental Nutrition Assistance Program (SNAP) - Key Statistics and Research,” www.ers.usda.gov, January 6, 2025, <https://www.ers.usda.gov/topics/food-nutrition-assistance/supplemental-nutrition-assistance-program-snap/key-statistics-and-research>.

^{vii} USDA Food and Nutrition Service, “Barriers That Constrain the Adequacy of SNAP Allotments,” June 2021, <https://fns-prod.azureedge.us/sites/default/files/resource-files/SNAP-Barriers-Summary.pdf>; Zoë Neuberger, Lauren Hall, and Linnea Sallack, “WIC’s Critical Benefits Reach Only Half of Those Eligible | Center on Budget and Policy Priorities,” Center on Budget and Policy Priorities, February 21, 2024, <https://www.cbpp.org/research/food-assistance/wics-critical-benefits-reach-only-half-of-those-eligible>.

^{viii} Matthew Rabbitt et al., “Household Food Security in the United States in 2022,” 2023, https://cps.ipums.org/cps/resources/food_security/Household%20Food%20Security%20in%20the%20United%20States%20in%202022.pdf.

^{ix} US Census Bureau, “Understanding and Using the American Community Survey Public Use Microdata Sample Files What Data Users Need to Know,” February 2021, https://www.census.gov/content/dam/Census/library/publications/2021/acs/acs_pums_handbook_2021.pdf.

^x Data accessed: US Census Bureau, “Food Security,” Census.gov, August 22, 2024, https://www.census.gov/data/datasets/time-series/demo/cps/cps-supp_cps-repwgt/cps-food-security.html#cpssupps.

^{xi} Data accessed here: US Census Bureau, “Accessing PUMS Data,” The United States Census Bureau, September 2024, <https://www.census.gov/programs-surveys/acs/microdata/access.html>.

^{xii} US Census Bureau, “Understanding and Using the American Community Survey Public Use Microdata Sample Files What Data Users Need to Know,” February 2021, https://www.census.gov/content/dam/Census/library/publications/2021/acs/acs_pums_handbook_2021.pdf.

^{xiii} US Census Bureau, “Understanding and Using the American Community Survey Public Use Microdata Sample Files What Data Users Need to Know,” February 2021,

https://www.census.gov/content/dam/Census/library/publications/2021/acs/acs_pums_handbook_2021.pdf.

^{xiv} Adam Shafi, “Sklearn Random Forest Classifiers in Python Tutorial,” www.datacamp.com, February 2023,

<https://www.datacamp.com/tutorial/random-forests-classifier-python>.

^{xv} Adam Shafi, “Sklearn Random Forest Classifiers in Python Tutorial,” www.datacamp.com, February 2023,

<https://www.datacamp.com/tutorial/random-forests-classifier-python>.